SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT TOOL 3 – Class Test

Course Code:	DSA03	**Course Title:**	Natural Language Processing
CO Assessed:	CO2 – Manipulation	**Assessment:**	Assessment Tool 3 – Class Test
Weightage:	20% of CIA		
CO5 Statement:	Design inflectional, derivational, finite-state parsing, stemming algorithms and for analyse morphological methods. (BL-3)		
Student Name:	D. Sri Anjaneya	**Reg. No.:**	192425219
Section / Batch:	AI & ML / 2024	**Date:**	03/08/2026

Code of Conduct

I D. Sri Anjaneya / 192425219 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _D. Sri Anjaneya_

Instructions

- Read all questions carefully before attempting the answers.
- Answer all five questions. Each question carries equal marks.
- Show all intermediate steps, calculations, probability computations, tagging decisions, and justifications wherever applicable.
- Duration: 30 minutes.

Section / Topic	Focus	Q Nos
N-gram Models and Word Counting for Language Modeling	N-gram Construction, Word Frequency Analysis, Probability Estimation	1
Unsmoothed N-grams and Smoothing Techniques in Syntax Analysis	Unsmoothed N-grams, Backoff, Deleted Interpolation	2
Entropy and Language Uncertainty in NLP Systems	Information Theory, Entropy Measurement, Language Prediction	3
English Word Classes and Tagsets for Linguistic Analysis	Word Categories, POS Categories, English Tagsets	4
Part-of-Speech Tagging Techniques and Applications	POS Tagging, Rule-Based Tagging, Stochastic Tagging, Transformation-Based Tagging	5

Annexure A – Class Test

1. A document processing system encounters the words “talked”, “talking”, and “talks”. Apply inflectional morphology rules and Porter Stemmer steps to derive the normalized root form. Identify the suffixes removed and explain the role of morphological normalization in improving information retrieval accuracy.

2. Given the input words: “relational”, “conditional”, “happiness”, “running”, design a hybrid morphological processing pipeline using derivational rules, inflectional analysis, finite-state morphological parsing, and Porter Stemmer. Decompose each word into root and affixes, represent transitions using finite-state parsing logic, and apply stemming steps to obtain the final normalized forms. Analyze how each technique contributes to accurate root extraction.

3. An enterprise-level search engine optimization (SEO) system processes web content containing variations such as “optimize”, “optimization”, “optimized”, and “optimizing”. The system must ensure consistent keyword indexing and ranking. Develop a morphology-driven solution integrating inflectional and derivational analysis, finite-state morphological parsing for rule validation, and Porter Stemmer for normalization. Apply the approach to the given inputs and determine how the system achieves improved search relevance and reduced redundancy.

4. A large-scale information retrieval system processes input words: “compute”, “computer”, “computing”, and “computation” to enhance search accuracy.
Design a hybrid morphological pipeline integrating inflectional and derivational analysis, finite-state morphological parsing, and Porter Stemmer.
Decompose each word into root and affixes, represent transformations using state transitions, and apply stemming rules for normalization.
Ensure consistent root extraction across all variations for efficient indexing.
Determine the morphological breakdown, parsing transitions, and final normalized forms for each word.

5.A text mining system processes input words: “decide”, “decision”, “decisive”, and “deciding” to support semantic clustering and analysis.
Construct a morphology-based framework combining derivational rules, inflectional transformations, finite-state parsing, and Porter stemming.
Analyze how suffix variations affect meaning and grammatical structure while maintaining root consistency. Apply the approach to extract roots, validate transformations, and normalize the dataset. Derive the morphological structure, state transitions, and final stem for each input word.

1. Inflectional Morphology and Porter Stemmer

Aim:

To normalize the words **talked, talking, talks** into their common root using inflectional morphology and the Porter Stemmer.

Procedure:

Step 1: Inflectional Morphology

Word	Suffix	Root
talked	-ed	talk
talking	-ing	talk
talks	-s	talk

Only grammatical suffixes are removed because the meaning remains unchanged.

Step 2: Porter Stemmer

Word	Porter Rule	Result
talked	Remove "ed"	talk
talking	Remove "ing"	talk
talks	Remove "s"	talk

Final Normalized Form:

Original Word	Final Stem
talked	talk
talking	talk
talks	talk

Observation:

- **-ed** indicates past tense.
- **-ing** indicates continuous tense.
- **-s** indicates third-person singular/plural.

All represent the same concept **talk**.

Role of Morphological Normalization:

- Reduces duplicate index entries.
- Improves search accuracy.
- Groups grammatical variants together.
- Saves storage.
- Improves Information Retrieval (IR).

Result:

All words are normalized into the single root: **talk**

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2. Hybrid Morphological Processing

Words:

- relational
- conditional
- happiness
- running

Step 1: Morphological Decomposition

Word	Root	Affix
relational	relate	-ional
conditional	condition	-al
happiness	happy	-ness
running	run	-ing

Step 2: Inflectional / Derivational Analysis

Word	Type
relational	Derivational
conditional	Derivational
happiness	Derivational
running	Inflectional

Step 3: Finite State Morphological Parsing

Example transitions

Relational:

Start

↓

relation

↓

+ al

↓

Accept

Conditional:

Start

↓

condition

↓

+ al

↓

Accept

Happiness:

Start

↓

happy

↓

+ ness

↓

Accept

Running:

Start

↓

run

↓

+ ing

↓

Accept

Step 4: Porter Stemmer

Word	**Rule**	**Stem**
relational	ational → ate	relat
conditional	tional → tion	condit
happiness	ness removed	happi
running	ing removed	run

Final Normalized Forms

Word	**Final Stem**
relational	relat
conditional	condit
happiness	happi
running	run

Analysis:

Derivational Rules

Change word meaning or category.

Example:

happy → happiness

Inflectional Rules:

Change only grammar.

Example:

run → running

Finite State Parser:

- Validates legal word formation.
- Recognizes suffixes.
- Ensures valid transitions.

Porter Stemmer:

Produces a common stem for indexing.

Result:

The hybrid pipeline extracts normalized roots suitable for Information Retrieval.

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3. SEO Morphology Pipeline

Words

- optimize
- optimization
- optimized
- optimizing

Morphological Breakdown

Word	**Root**	**Affix**
optimize	optimize	—
optimization	optimize	-ation
optimized	optimize	-ed
optimizing	optimize	-ing

Inflectional / Derivational

Word	Type
optimize	Base
optimization	Derivational
optimized	Inflectional
optimizing	Inflectional

Finite State Parsing:

1.Start

↓

optimize

↓

+ ation

↓

Accept

2.Start

↓

optimize

↓

+ ed

↓

Accept

3.Start

↓

optimize

↓

+ ing

↓

Accept

Porter Stemmer

Word	Stem
optimize	optim
optimization	optim
optimized	optim
optimizing	optim

Final Normalized Form:

All words become

optim

SEO Benefits

- Single keyword index
- Better ranking
- Reduced redundancy
- Improved query matching
- Faster retrieval

Result:

All variants are indexed under one stem: **optim**

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4. Hybrid Morphology for Compute

Words

- compute
- computer
- computing
- computation

Morphological Breakdown:

Word	Root	Affix
compute	compute	—
computer	compute	-er
computing	compute	-ing
computation	compute	-ation

Type:

Word	Type
compute	Base
computer	Derivational
computing	Inflectional
computation	Derivational

Finite State Parsing:

Computer:

Start

↓

compute

↓

+ er

↓

Accept

Computing:

Start

↓

compute

↓

+ ing

↓

Accept

Computation:

Start

↓

compute

↓

+ ation

↓

Accept

Porter Stemmer:

Word	**Stem**
compute	comput
computer	comput
computing	comput
computation	comput

Final Normalized Form:

comput

Benefits:

- Same index entry
- Better search
- Less storage
- Improved Information Retrieval

Result:

All words normalize to

Comput

5. Morphology Framework for Decide

Words

- decide
- decision
- decisive
- deciding

Morphological Breakdown

Word	**Root**	**Affix**
decide	decide	—
decision	decide	-sion
decisive	decide	-ive
deciding	decide	-ing

Type:

Word	**Type**
decide	Base
decision	Derivational
decisive	Derivational
deciding	Inflectional

Finite State Parsing:

Decision:

Start

↓

decide

↓

+ sion

↓

Accept

Decisive:

Start

↓

decide

↓

+ ive

↓

Accept

Deciding:

Start

↓

decide

↓

+ ing

↓

Accept

Porter Stemmer

Word	**Stem**
decide	decid
decision	decis
decisive	decis
deciding	decid

Meaning Analysis

Word	**Meaning**
decide	Verb
decision	Noun
decisive	Adjective
deciding	Verb (continuous form)

Derivational suffixes change the grammatical category, whereas inflectional suffixes only change the tense or grammatical form while preserving the core meaning.

Framework Contribution

- Derivational analysis identifies changes in meaning and word class.
- Inflectional analysis removes grammatical endings.
- Finite-state parsing validates legal morphological structures.
- Porter stemming normalizes related words for indexing and clustering.

Result:

The normalized stems are:

Word	Final Stem
decide	decid
decision	decis
decisive	decis
deciding	decid

Rubrics for Calculation

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Task	20%	Clearly understands the given task/experiment without assistance	Understands task with minor clarification	Partial understanding; requires guidance	Unable to understand the task
Procedure & Execution	25%	Performs procedure accurately and independently with proper technique	Performs with minor errors	Requires guidance; makes procedural mistakes	Unable to execute the procedure
Observation & Result	20%	Records accurate observations and obtains correct results	Minor errors in observation or result	Incomplete or partially correct results	Incorrect or no results
Viva Voce / Conceptual Response	20%	Answers all questions confidently with strong conceptual clarity	Answers most questions with minor gaps	Limited responses; needs prompting	Unable to answer questions
Time Management & Presentation	15%	Completes task on time with neat and clear presentation	Completes with minor delay or presentation issues	Delayed or poorly presented work	Unable to complete on time

Q. No. / Criteria Level	Understanding of Task	Procedure & Execution	Observation & Result	Viva Voce / Conceptual Response	Time Management & Presentation	Sub. Total	Total %
1							

2							
3							
4							
5							

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____